

MATH0118 Mathematics for Quantum Mechanics

<i>Year:</i>	2026–2027
<i>Code:</i>	MATH0118
<i>Level:</i>	7 (UG), 7 (PG)
<i>Normal student group(s):</i>	UG: Y3, Y4 Mathematics programmes PG: MSc Mathematics
<i>Value:</i>	15 credits (7.5 ECTS credits)
<i>Term:</i>	1
<i>Assessment:</i>	80% examination, 20% coursework
<i>Normal Pre-requisites:</i>	MATH0056 (Mathematical Methods 4) MATH0014 (Algebra 3: Further Linear Algebra)
<i>Lecturer:</i>	Prof R Halburd

Course Description and Objectives

This module aims to introduce students to the mathematical framework used in quantum mechanics and teach them to apply various approximation methods, with minimal prior knowledge of quantum physics. As well as underpinning much of modern physics, quantum mechanics is important to mathematics, both as a source of important applications and as a source of ideas that have led to new directions in pure mathematics. After a brief overview of the physical motivation for quantum mechanics, we will introduce the main postulates and mathematical framework. We will solve the Schrödinger equation exactly in several important cases, such as the single-electron atom, and introduce the idea of internal symmetries and spin. Various approximation methods will be studied, including first- and second-order perturbation theory and the Wentzel, Kramers, Brillouin (WKB) approximation method.

Recommended Texts

- S Weinberg. *Lectures on Quantum Mechanics* 2nd ed (Cambridge)
- JJ Sakurai and J Napolitano. *Modern Quantum Mechanics* 2nd ed (Addison–Wesley)
- E Merzbacher. *Quantum Mechanics* 3rd ed (Wiley)
- SJ Gustafson and IM Sigal, *Mathematical Concepts of Quantum Mechanics* 3rd ed (Springer)

Detailed Syllabus

Key experiments. Wave mechanics. Matrix mechanics. Probabilistic interpretation. Operators on vector spaces. Position and momentum operators. The Schrödinger equation. The harmonic oscillator. Creation and annihilation operators. Particle in a central potential. Spherical harmonics. General principles of quantum mechanics. Observables. Symmetries. Heisenberg uncertainty principle. Rotations, angular momentum and spin. Tensor products of representations. Addition of angular momentum. Clebsch-Gordan coefficients. Bosons, fermions and internal symmetries. The algebraic derivation of the hydrogen spectrum. Time-independent approximations. First-order perturbation theory. The Zeeman effect. The Stark effect. Second-order perturbation theory. The variational method. The Born-Oppenheimer approximation. The Wentzel, Kramers, Brillouin (WKB) approximation. Time-dependent approximations. Monochromatic perturbations. Absorption and emission of radiation. Scattering theory. Resonances.

September 2026